

Monday

WTW158 Calculus

8

Class Test 1 Mo

27 February - 3 March 2023

Time: 15 min

Surname:

Initials:

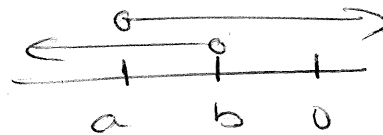
Student number:

Group:

Use a pen to complete the test. Sketches must also be in pen. Simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks]

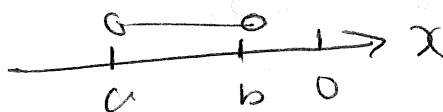
Let $a < b < 0$. Suppose $x < b$ and $x > a$.



(i) Write down the interval on which x lies. Simplify your answer.

$$x \in (a, b)$$

(ii) Indicate your final answer in (i) on a number line.



QUESTION 2 [2 marks]

Simplify the following completely.

If $x^2 = \pi^2$, then $x = \pm \pi$ and $\sqrt{x^2} = \pi$

QUESTION 3 [1 mark]

Circle all the rational numbers in the table below. If you circle a number that is not rational, you will get zero for the question.

1.6	$\frac{\pi}{2}$	3.145
-----	-----------------	-------

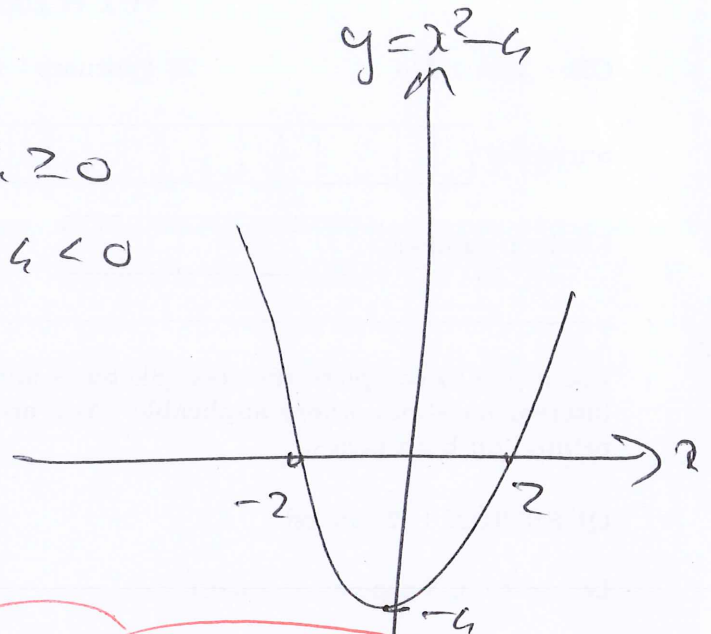
If $\frac{\pi}{2}$ was circled: 0 marks.

QUESTION 4 [3 marks]

Show how you use the definition of the absolute value to sketch the graph of $y = |x^2 - 4|$. Indicate all relevant detail, e.g., intercepts, on your graph.

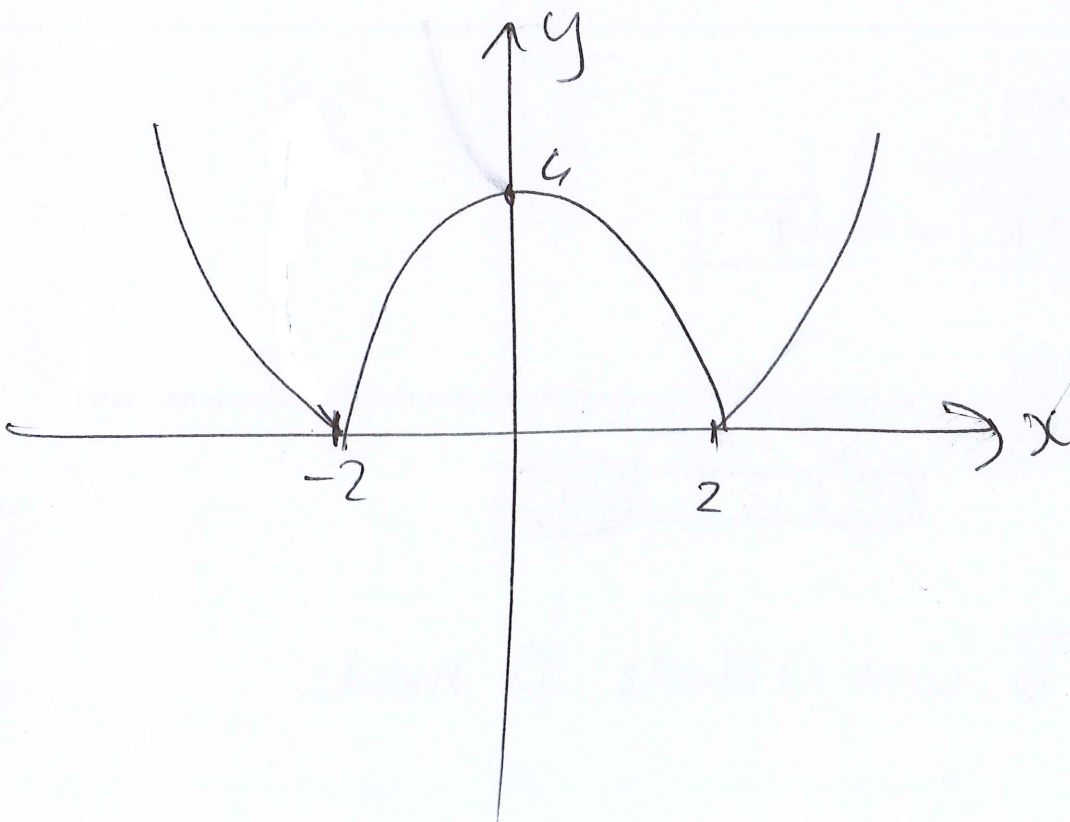
$$y = |x^2 - 4|$$

$$= \begin{cases} x^2 - 4 & \text{if } x^2 - 4 \geq 0 \\ -(x^2 - 4) & \text{if } x^2 - 4 < 0 \end{cases}$$



$$= \begin{cases} x^2 - 4 & \text{if } x \in (-\infty, -2] \cup [2, \infty) \\ -(x^2 - 4) & \text{if } x \in (-2, 2) \end{cases}$$

This you obtain from the graph



Tuesday

WTW158 Calculus

Class Test 1 (Tu)

27 February - 3 March 2023

Time: 15 min

8

Surname:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Initials:

--	--	--

Student number:

--	--	--	--	--	--	--	--

Group:

--	--	--

Use a pen to complete the test. Sketches must also be in pen. Simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

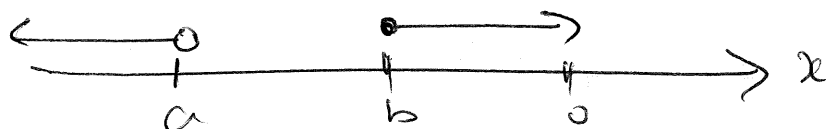
QUESTION 1 [2 marks]

Let $a < b < 0$. Suppose $x < a$ or $x \geq b$.

(i) Write down the interval in which x lies. Simplify your answer.

$$x \in (-\infty, a) \cup [b, \infty)$$

(ii) Indicate your final answer in (i) on a number line.



QUESTION 2 [2 marks]

Simplify the following. Write your answers without the absolute value sign.

If $a < b < 0$, then $|ab| = ab$ and $\sqrt{a^2} = -a$

QUESTION 3 [1 mark]

Circle all the rational numbers in the table below. If you circle a number that is not rational, you will get zero for the question.

3.145	$\frac{22}{7}$	$\frac{2}{\sqrt{2}}$
-------	----------------	----------------------

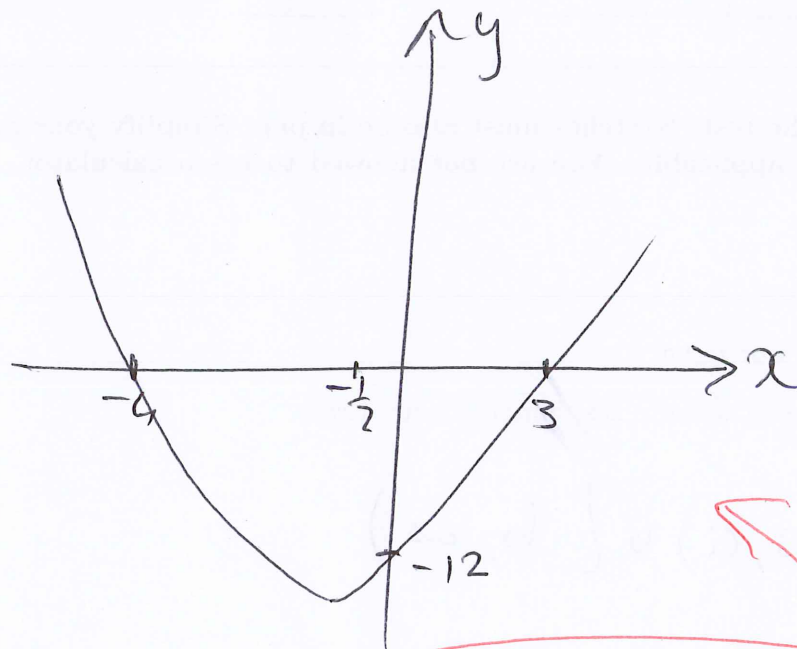
If $\frac{2}{\sqrt{2}}$ is circled: 0 marks.

QUESTION 4 [3 marks]

Show how you use the definition of the absolute value to sketch the graph of $y = |x^2 + x - 12|$. Indicate all relevant detail, e.g., intercepts, on your graph.

$$y = |x^2 + x - 12|$$

$$= \begin{cases} x^2 + x - 12 & \text{if } x^2 + x - 12 \geq 0 \\ -(x^2 + x - 12) & \text{if } x^2 + x - 12 < 0 \end{cases}$$

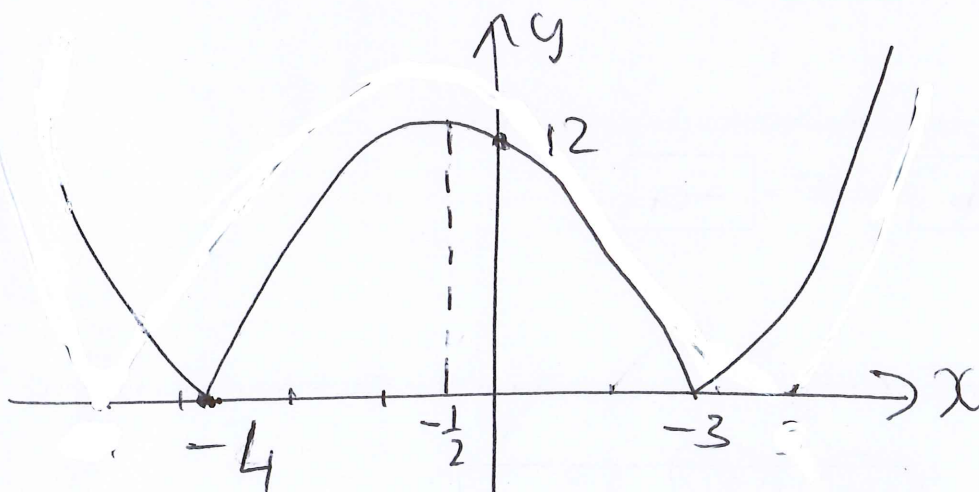


$$x^2 + x - 12 = (x+4)(x-3)$$

$$\frac{-b}{2a} = -\frac{1}{2}$$

from graph.

$$= \begin{cases} x^2 + x - 12 & \text{if } x \in (-\infty, -4] \cup [3, \infty) \\ -(x^2 + x - 12) & \text{if } x \in (-4, 3) \end{cases}$$



All details to be given on graph.

Thursday

WTW158 Calculus

8

Class Test 1 (Th)

27 February - 3 March 2023

Time: 15 min

Surname:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Initials:

--	--	--

Student number:

--	--	--	--	--	--	--	--

Group:

--	--	--

Use a pen to complete the test. Sketches must also be in pen. Simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

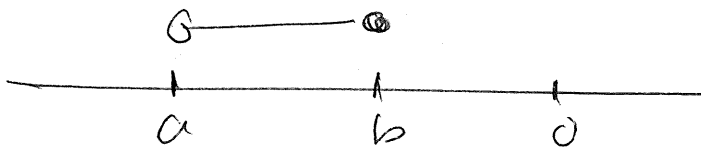
QUESTION 1 [2 marks]

Let $a < b < 0$ and suppose $x > a$ and $x \leq b$.

(i) Write down the interval on which x lies. Simplify your answer.

$$x \in (a, b]$$

(ii) Indicate your final answer in (i) on a number line.



QUESTION 2 [2 marks]

Simplify the following. Write your answers without the absolute value sign.

If $a < 0 < b$, then $|ab| = \boxed{-ab}$ and $\sqrt{b^2} = \boxed{b}$

QUESTION 3 [1 mark]

Circle all the rational numbers in the table below. If you circle a number that is not rational, you will get zero for the question.

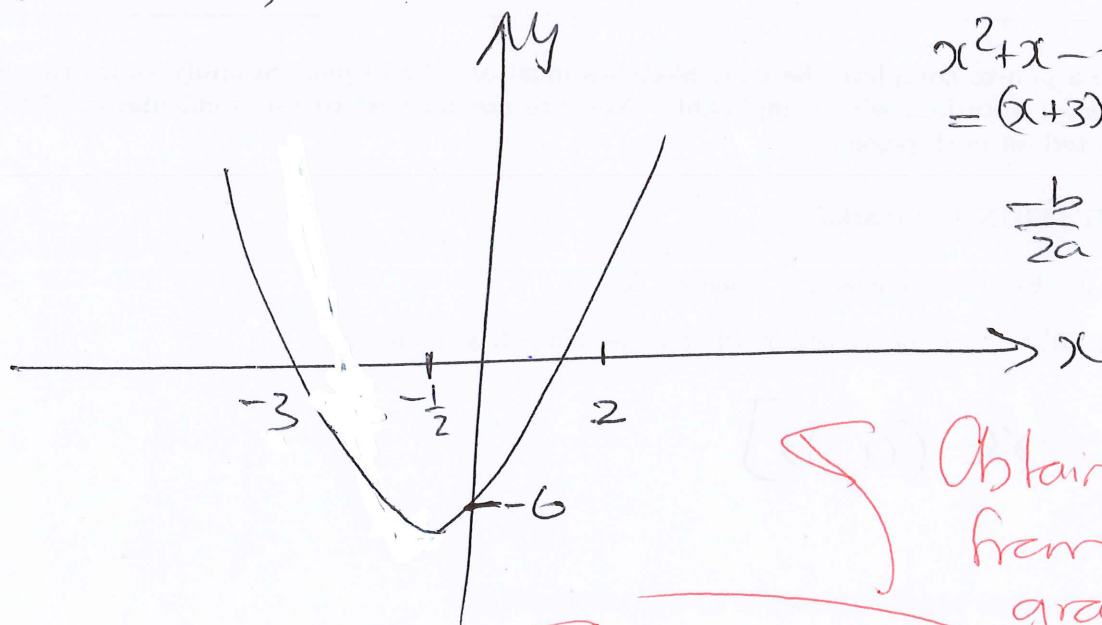
3.145	$\frac{22}{7}$	$\sqrt{49}$
-------	----------------	-------------

1

QUESTION 4 [3 marks]

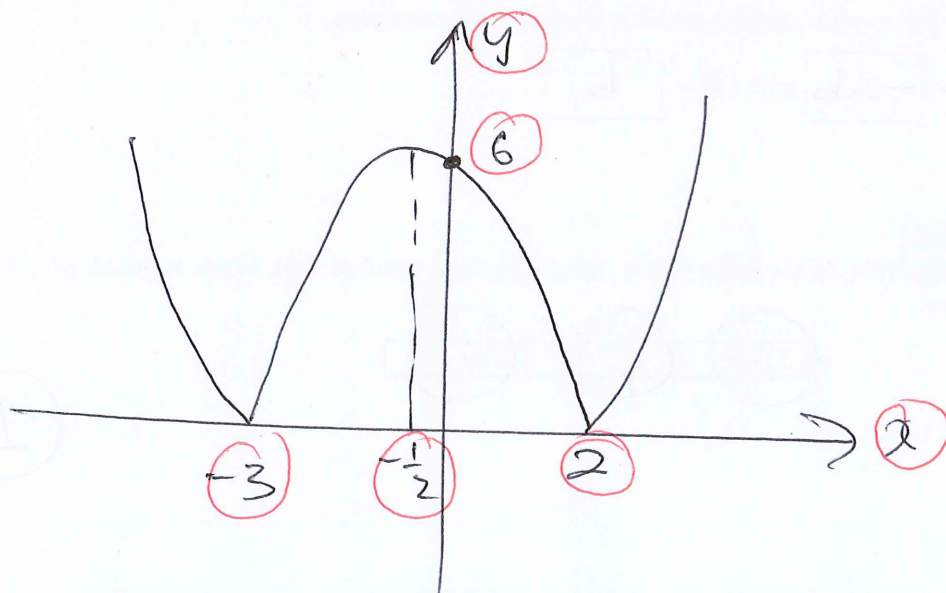
Show how you use the definition of the absolute value to sketch the graph of $y = |x^2 + x - 6|$. Indicate all relevant detail, e.g., intercepts, on your graph.

$$y = |x^2 + x - 6| = \begin{cases} x^2 + x - 6 & \text{if } x^2 + x - 6 \geq 0 \\ -(x^2 + x - 6) & \text{if } x^2 + x - 6 < 0 \end{cases}$$



Obtain from graph.

$$= \begin{cases} x^2 + x - 6 & \text{if } x \in (-\infty, -3] \cup [2, \infty) \\ -(x^2 + x - 6) & \text{if } x \in (-3, 2) \end{cases}$$



Friday

WTW158 Calculus

Class Test 1 (Fr)

27 February - 3 March 2023

Time: 15 min

8

Surname:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Initials:

--	--	--

Student number:

--	--	--	--	--	--	--	--

Group:

--	--	--

Use a pen to complete the test. Sketches must also be in pen. Simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

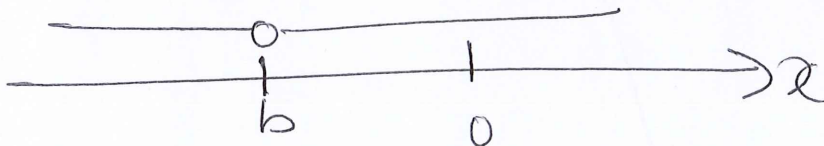
QUESTION 1 [2 marks]

Consider all numbers x such that $x < b$ or $x > b$.

(i) Write down the interval on which x lies. Simplify your answer.

$$x \in (-\infty, b) \cup (b, \infty)$$

(ii) Indicate your final answer in (i) on a number line if $b < 0$.



QUESTION 2 [2 marks]

Simplify the following. Write your answers without the absolute value sign.

If $b < 0 < a$, then $|ab| = \boxed{-ab}$ and $\sqrt{a^2} = \boxed{a}$

QUESTION 3 [1 mark]

Circle all the rational numbers in the table below. If you circle a number that is not rational, you will get zero for the question.

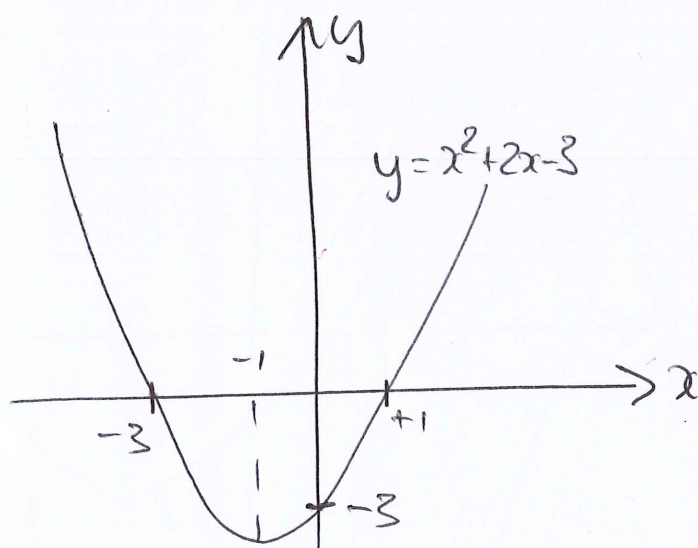
1.3	$\frac{\sqrt{64}}{3}$	$\frac{1.8}{3}$
-----	-----------------------	-----------------

QUESTION 4 [3 marks]

Show how you use the definition of the absolute value to sketch the graph of $y = |x^2 + 2x - 3|$. Indicate all relevant detail, e.g., intercepts, on your graph.

$$y = |x^2 + 2x - 3|$$

$$= \begin{cases} x^2 + 2x - 3 & \text{if } x^2 + 2x - 3 \geq 0 \\ -(x^2 + 2x - 3) & \text{if } x^2 + 2x - 3 < 0 \end{cases}$$



$$x^2 + 2x - 3 = (x+3)(x-1)$$

from this graph

$$= \begin{cases} x^2 + 2x - 3 & \text{if } x \in (-\infty, -3] \cup [1, \infty) \\ -(x^2 + 2x - 3) & \text{if } x \in (-3, 1) \end{cases}$$

